F-01 | Native microscopy fluorescence analysis

How does recorded fluorescence change across the supplied cadmium-labelled microscopy acquisitions?

Materials and methods

116 original supplied files and 4 derived context files; ND2, TIFF, AVI and experimental notes; original intensities preserved.

nd2 0.12.0; tifffile 2026.10.15; scikit-image 0.26.0; SciPy 1.18.1.

Stream native frames; distinguish T,C,P,Z axes; cross-check ND2 timestamps against per-frame metadata. Use acquired event Z coordinates and cross-check optical stacks against Z index and planned positions.

Whole-field native means are primary. Estimate spatial background outside BF-derived regions; keep raw means and a 10th-percentile background sensitivity.

BF high-pass regions use a fixed initial threshold, 1 µm expansion and 4-150 µm² area limits. Regions are not validated cells.

Initial/final 10% medians; fractional changes only above 5× spatial noise. Fixed-mask and 0.85/1.15 threshold sensitivity; every-frame clipping, focus and drift.

Retain all frames. AVI is secondary export QC; old Z stacks and static fields are not time courses.

Scope and working assumptions

Conditions are nominal: 1 mM Cd stock contradicts the 5 mM labels. The C2 Ca 0/1 mM filename conflict is retained. Initial reference is recording-relative, not a verified pre-Cd baseline. TIFF clocks conflict; frame-index analysis only. Denoise.ai ON means acquisition-processed data, not unprocessed detector counts. BF-region masks do not establish single-cell identities; dense 30 September masks are incomplete. Native time-series exposure event values are missing and are not imputed from static-image 10 ms metadata. Stage Z varies, but focus compensation can move the objective; PFS state codes are opaque here. Pairs remain conditional on optical comparability. C1/Empty pairs change both Ca and Cd, so they do not isolate Cd. Fields/cells/frames/exports are not independent cultures. No automatic photobleach correction, culture p-values, binding or calibrated dose model.

Results

120 input files, including derived context, were inventoried and byte-verified; 107 native/export image containers were processed.

20 time-series acquisitions; 14 with native clocks. All 146 static/Z frames are retained; 141 contain fluorescence and 94 have a usable BF-region metric.

13 October, nominal 10 mM Ca / 5 mM Cd: C8: +2.66 ADU over 298.2s; Empty: +0.12 ADU over 299.5s; C6: +15.01 ADU over 298.4s; C1: +0.57 ADU over 299.6s; C2: +0.77 ADU over 299.6s. The C6 recording has the largest within-recording increase in this group. Candidate field-matched pre/post contrasts are C1 -10.22 ADU; C2 +1.92 ADU; Empty +0.68 ADU. Within-recording change and the field-matched treatment contrast answer different questions: the C1 field decreases across its paired acquisitions despite rising slightly within the challenged recording.

No recording met the preset resolved-plateau criterion; an empirical reaction time constant, t50 and t90 were not estimated.

Four same-well comparisons meet the stage/BF field-match criterion. Their pre/post contrasts are tabulated separately from within-recording change.

Recorded stage Z spans 2.34 µm in C2 Ca 1 mM / Cd 5 mM and 1.08 µm in C2 Ca 10 mM / Cd 5 mM. PFS Status is code 1 except for three code-5 frames across those two C2 recordings; PFS offsets are constant within all 14 native time series. Codes are retained without assigning a lock-state meaning; stage motion alone does not establish defocus.

6 AVI files were decoded as processed-export QC and were not counted as additional biological observations.

Record	Input data	Final − initial (ADU)	Time coverage	Finished without more wet-lab data
R020_P0_Z0	F01_13Oct_C6_10mM-calcium_500uM-cadmium_D7_t5min_1001.nd2	+0.05	297.6s	Yes: descriptive recording analysis
R021_P0_Z0	F01_13Oct_C8_10mM-calcium_5mM-cadmium_E6_t5min_1001.nd2	+2.66	298.2s	Yes: descriptive recording analysis
R022_P0_Z0	F01_13Oct_Empty_10mM-calcium_5mM-cadmium_A6_t5min_1001.nd2	+0.12	299.5s	Yes: descriptive recording analysis
R023_P0_Z0	F01_13Oct_C6_10mM-calcium_5mM-cadmium_D6_t5min_1001.nd2	+15.01	298.4s	Yes: descriptive recording analysis
R024_P0_Z0	F01_13Oct_C1_10mM-calcium_5mM-cadmium_B6_t5min_1001.nd2	+0.57	299.6s	Yes: descriptive recording analysis
R025_P0_Z0	F01_13Oct_C2_10mM-calcium_5mM-cadmium_B7_t5min_1001.nd2	+0.77	299.6s	Yes: descriptive recording analysis
R026_P0_Z0	F01_13Oct_C2_1mM-calcium_5mM-cadmium_B6_t5min_1.nd2	+2.40	299.3s	Yes: descriptive recording analysis
R027_P0_Z0	F01_13Oct_C2_10mM-calcium_500uM-cadmium_B5_t5min_1.nd2	+0.26	301.4s	Yes: descriptive recording analysis
R028_P0_Z0	F01_13Oct_C1_1mM-calcium_0mM_cadmium_B5_t0_1001.nd2	-0.50	304.0s	Yes: descriptive recording analysis
R041_P0_Z0	F01_30Sep_E5_Colony6_calcium-1mM_FLUO_cadmium_timelapse10s-240s.nd2	-0.55	120.0s	Yes: descriptive recording analysis
R042_P0_Z0	F01_30Sep_E5_Colony6_calcium-1mM_FLUO_cadmium_timelapse10s-240s002.nd2	+0.29	120.0s	Yes: descriptive recording analysis
R043_P0_Z0	F01_30Sep_E5_Colony6_calcium-1mM_FLUO_cadmium_timelapse10s-240s003.nd2	-0.52	360.0s	Yes: descriptive recording analysis
R044_P0_Z0	F01_30Sep_C5_Colony6_no-calcium_FLUO_cadmium_timelapse5s-300s-001.nd2	-0.05	300.0s	Yes: descriptive recording analysis

Record	Input data	Final − initial (ADU)	Time coverage	Finished without more wet-lab data
R045_P0_Z0	F01_30Sep_E6_Colony8_calcium-1mM_FLUO_cadmium_timelapse5s-300s-001.nd2	+1.26	300.0s	Yes: descriptive recording analysis
R058_P0_Z0	F01_09Oct_Control_no_Cadmium__B3Ca10mMBaselinecontrol.nd2_-_B3Ca10mMBaselinecontrol.nd2_series_1_.tif	+0.29	frame-index only	Yes: descriptive recording analysis
R059_P0_Z0	F01_09Oct_Control_no_Cadmium__B3Ca10mMBaselinecontrol.nd2_-_B3Ca10mMBaselinecontrol.nd2_series_2_.tif	-0.59	frame-index only	Yes: descriptive recording analysis
R060_P0_Z0	F01_09Oct_Control_no_Cadmium__B3Ca10mMBaselinecontrol.nd2_-_B3Ca10mMBaselinecontrol.nd2_series_3_.tif	+1.61	frame-index only	Yes: descriptive recording analysis
R061_P0_Z0	F01_09Oct_Cd5mM-Sample_S1_B1Cd5mM001.nd2_-_B1Cd5mM001.nd2_series_1_.tif	+4.53	frame-index only	Yes: descriptive recording analysis
R062_P0_Z0	F01_09Oct_Cd5mM-Sample_S1_B1Cd5mM001.nd2_-_B1Cd5mM001.nd2_series_3_.tif	+8.04	frame-index only	Yes: descriptive recording analysis
R063_P0_Z0	F01_09Oct_Cd5mM-Sample_S1_B1Cd5mM001.nd2_-_B1Cd5mM001.nd2_series_2_.tif	+9.46	frame-index only	Yes: descriptive recording analysis

Interpretation

13 October, nominal 10 mM Ca / 5 mM Cd: C8: +2.66 ADU over 298.2s; Empty: +0.12 ADU over 299.5s; C6: +15.01 ADU over 298.4s; C1: +0.57 ADU over 299.6s; C2: +0.77 ADU over 299.6s. The C6 recording has the largest within-recording increase in this group. Candidate field-matched pre/post contrasts are C1 -10.22 ADU; C2 +1.92 ADU; Empty +0.68 ADU. Within-recording change and the field-matched treatment contrast answer different questions: the C1 field decreases across its paired acquisitions despite rising slightly within the challenged recording.

References and reproducibility

Chen et al. (2013), Ultrasensitive fluorescent proteins for imaging neuronal activity. https://doi.org/10.1038/nature12354

Montero Llopis et al. (2021), Best practices and tools for reporting reproducible fluorescence microscopy methods. https://doi.org/10.1038/s41592-021-01156-w

nd2 official documentation: https://talleylambert.com/nd2/

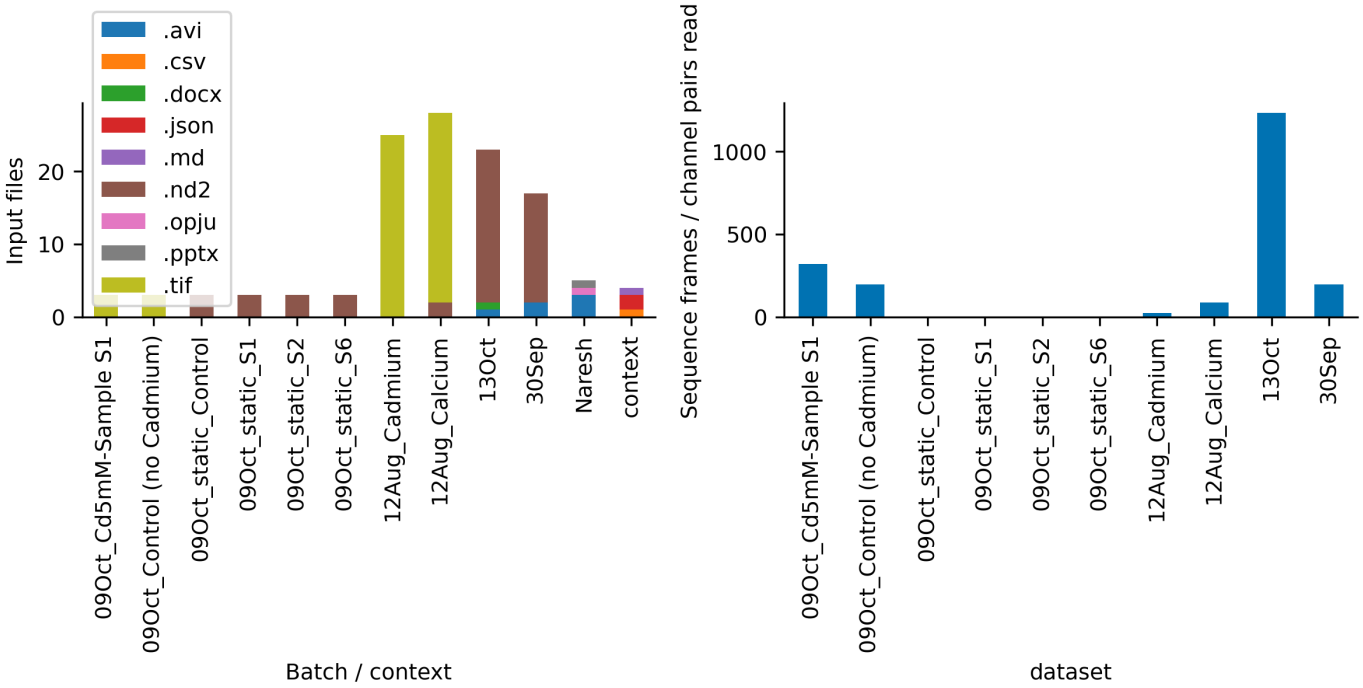
scikit-image registration documentation: https://scikit-image.org/docs/stable/api/skimage.registration.html

Nikon Perfect Focus System: near-infrared focus compensation maintains objective-to-coverslip position. https://www.microscope.healthcare.nikon.com/en_EU/resources/glossary/perfect-focus-system

Supplied 13 October lab notebook, body.b0029 (condition table), b0037-b0069 (prior qualitative observations).

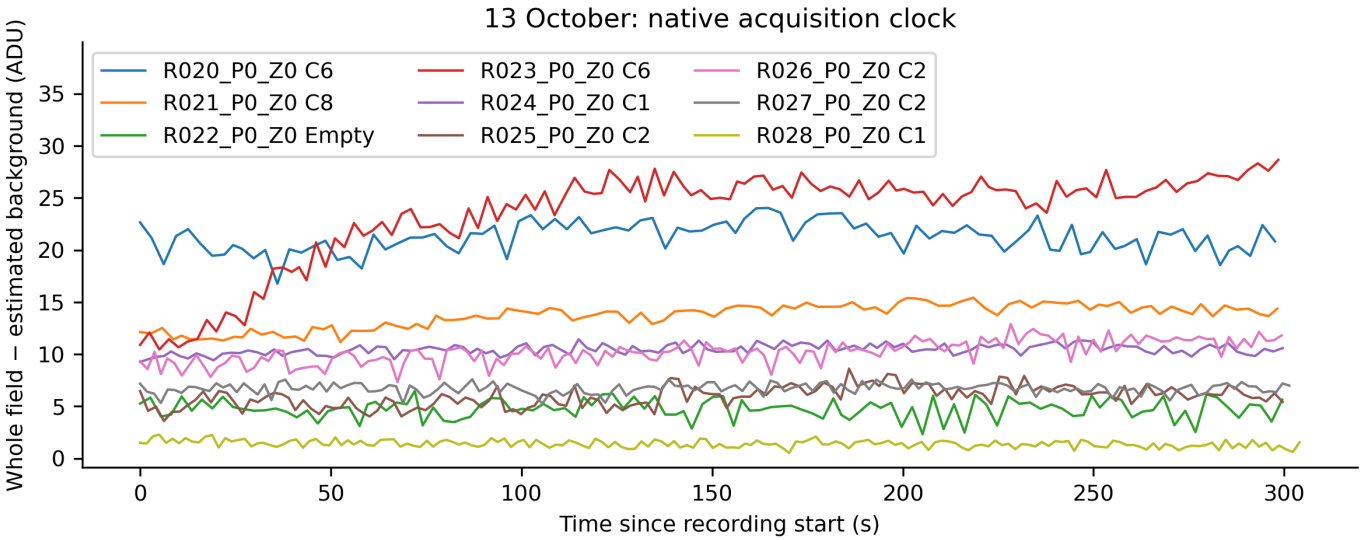
Biomni methods review supplied in F01_biomni_methods_review.json; no arbitrary drift cutoff adopted.

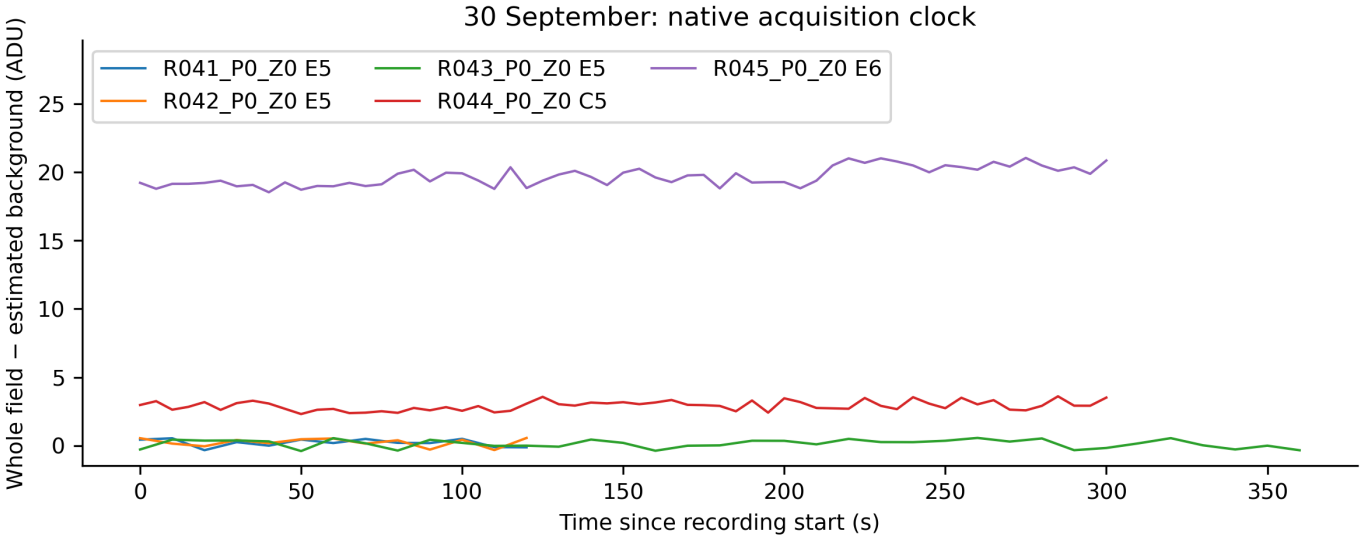
Figure 1



Input files and sequence frames or channel pairs. Native and exported files remain separately identified; bars are not biological replicate counts.

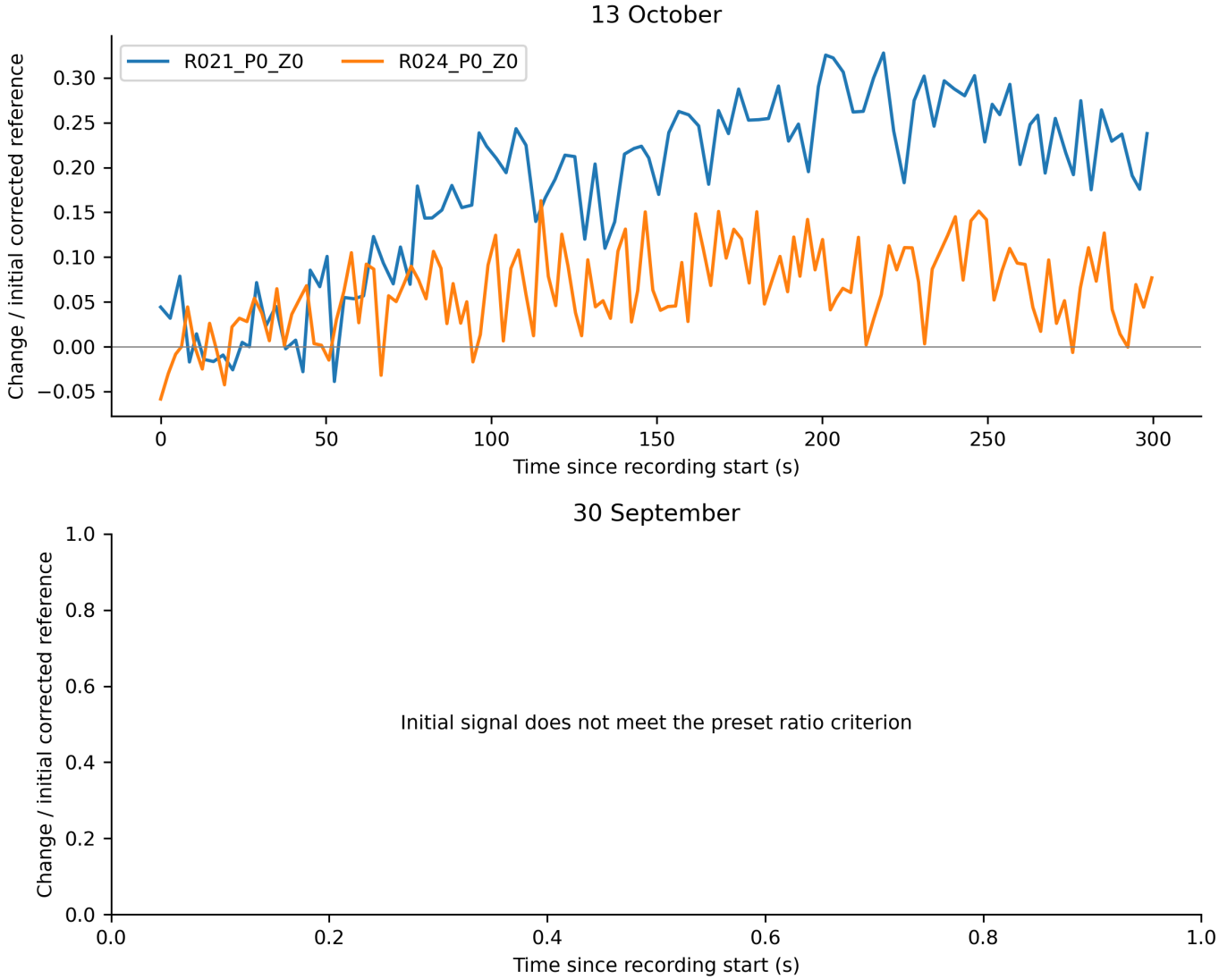
Figure 2





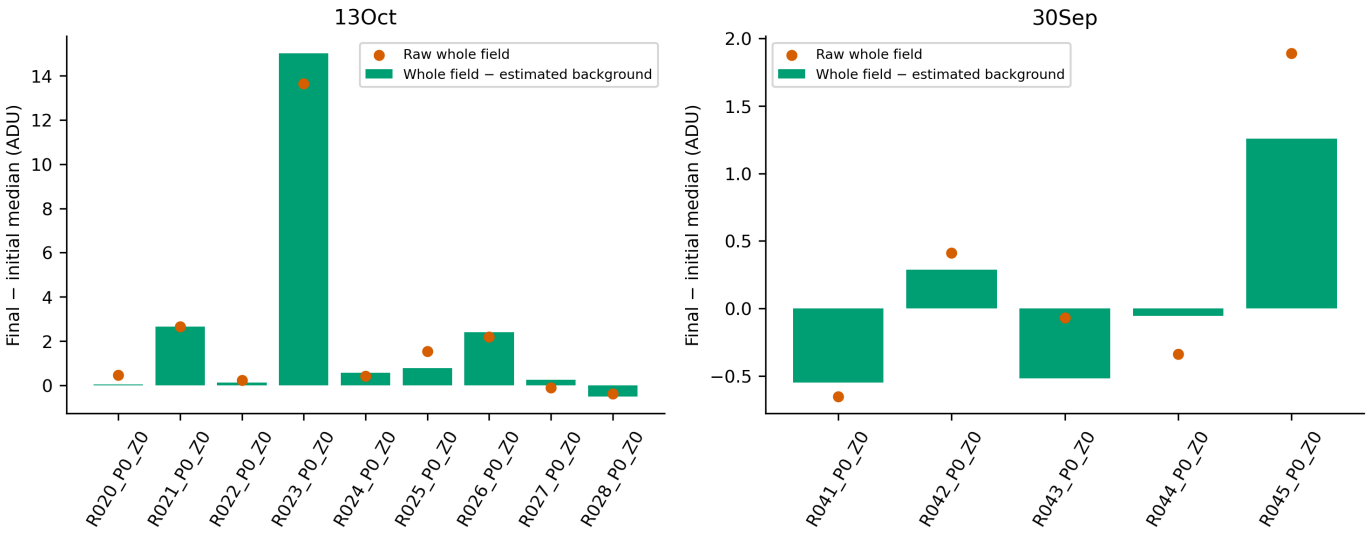
All native time-series traces; references and positions belong to individual recordings. No Cd-addition frame is assumed.

Figure 3



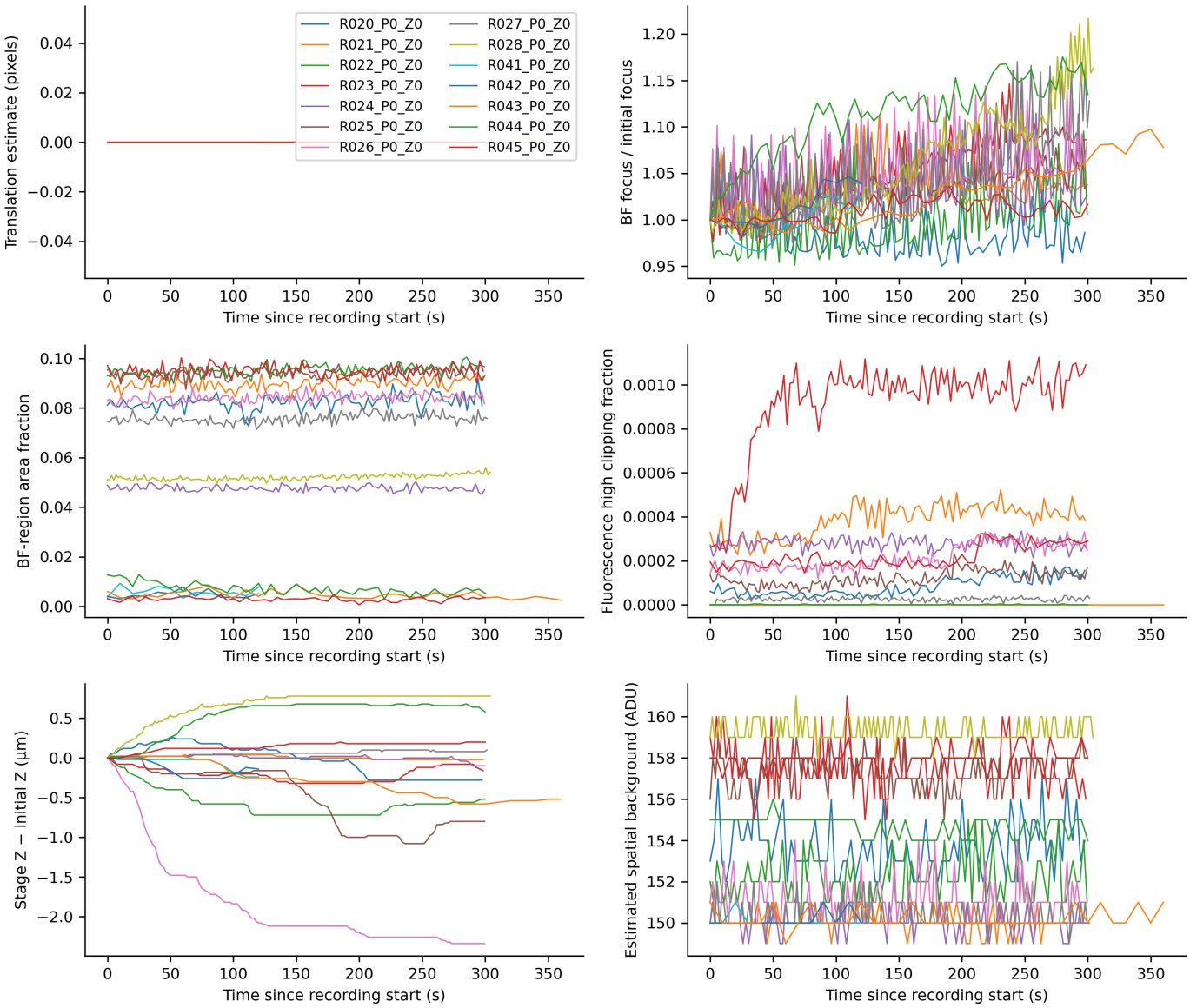
Fractional changes use the first 10% of each recording, not a pre-exposure baseline. Ratios are omitted when the reference is too small relative to spatial background noise.

Figure 4



Acquisition-level whole-field changes with spatial background subtraction (bars) and untouched raw whole-field means (points). BF-region changes and fixed-mask sensitivity remain in the numerical tables.

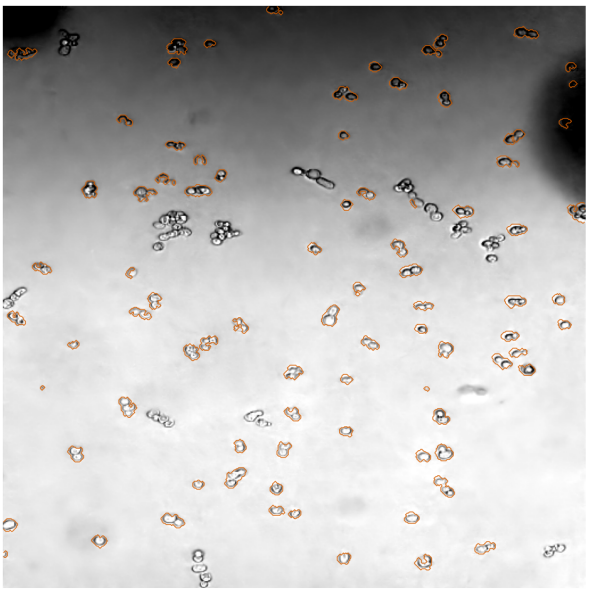
Figure 5



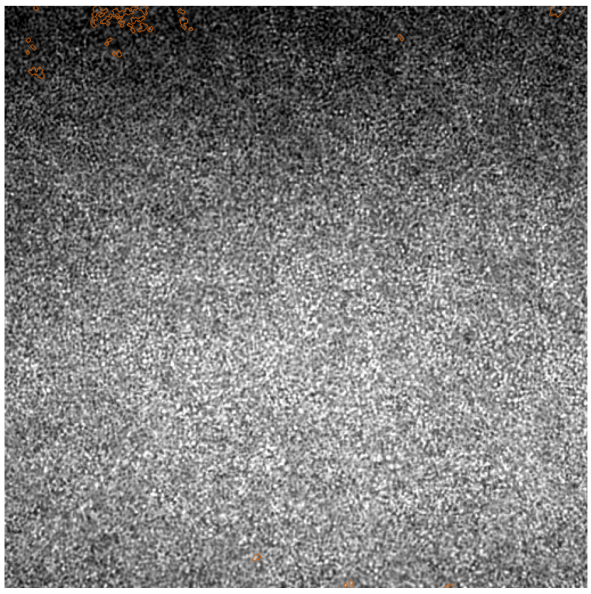
Every-frame translation, transmitted-light focus, mask coverage, clipping, recorded stage Z and spatial background. Stage motion alone does not establish defocus; recorded PFS codes/offsets accompany the Z trajectories in the QC tables. No automatic exclusion or bleaching correction was applied.

Figure 6

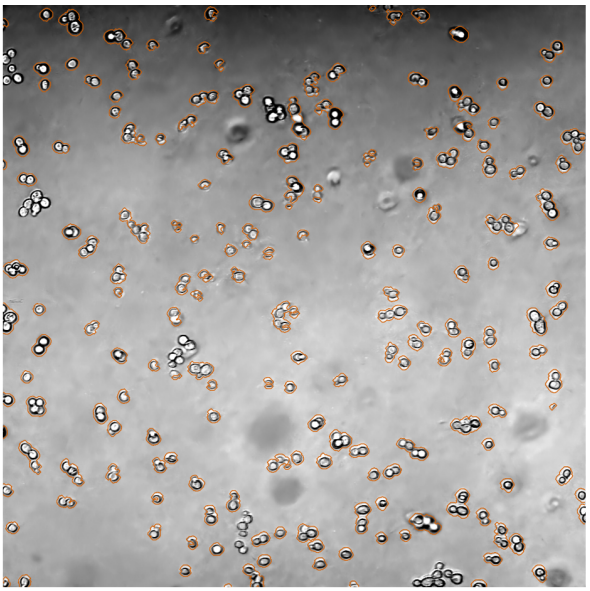
13Oct R007



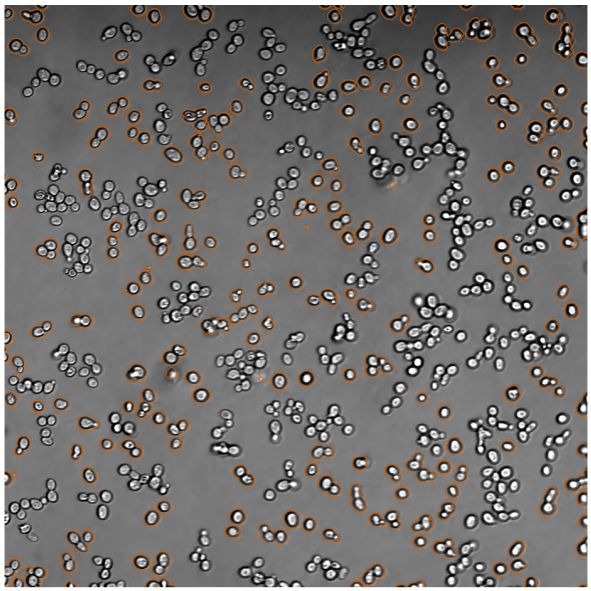
30Sep R031



09Oct_static_Control R046

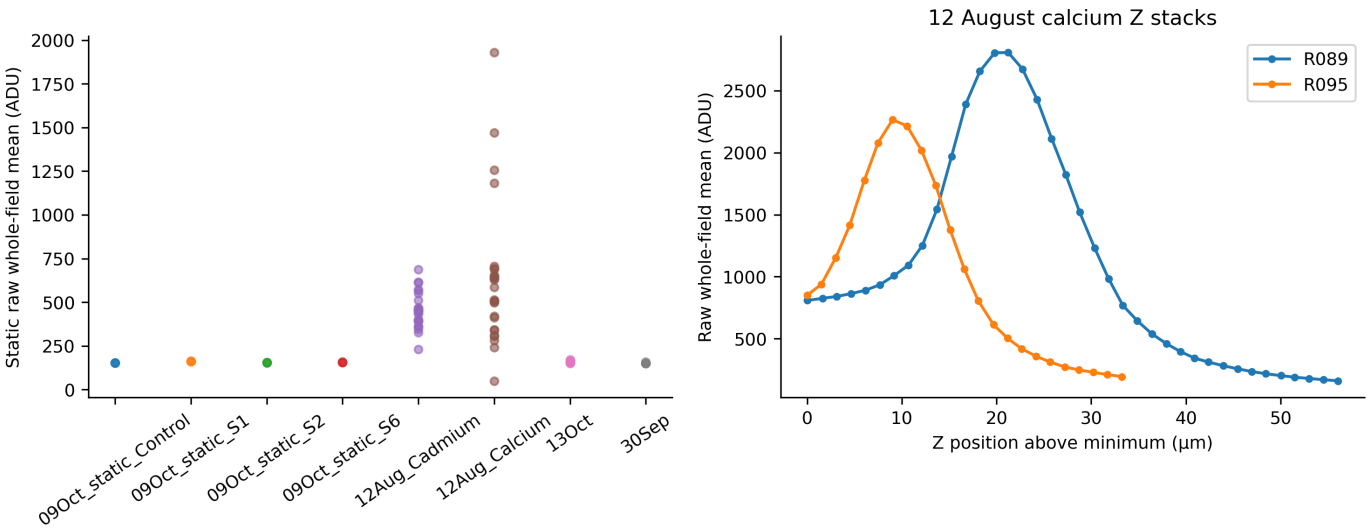


09Oct_static_S1 R049



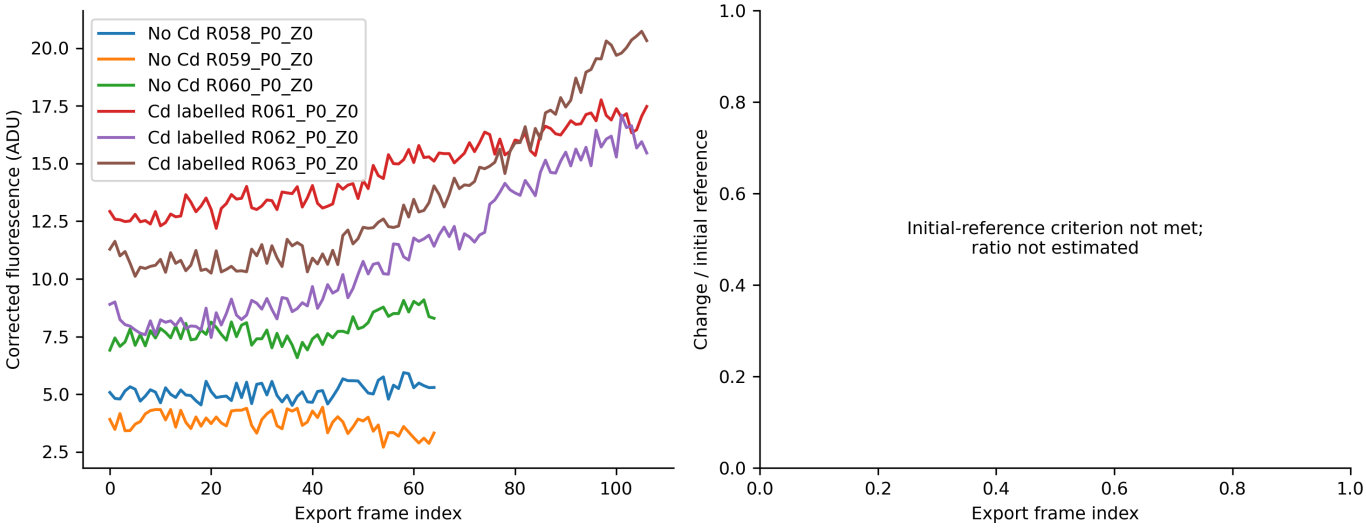
Representative BF-only region masks. First, middle, last and at least 5% of frame overlays are supplied in quality_control/. Region counts are not certified cell counts.

Figure 7



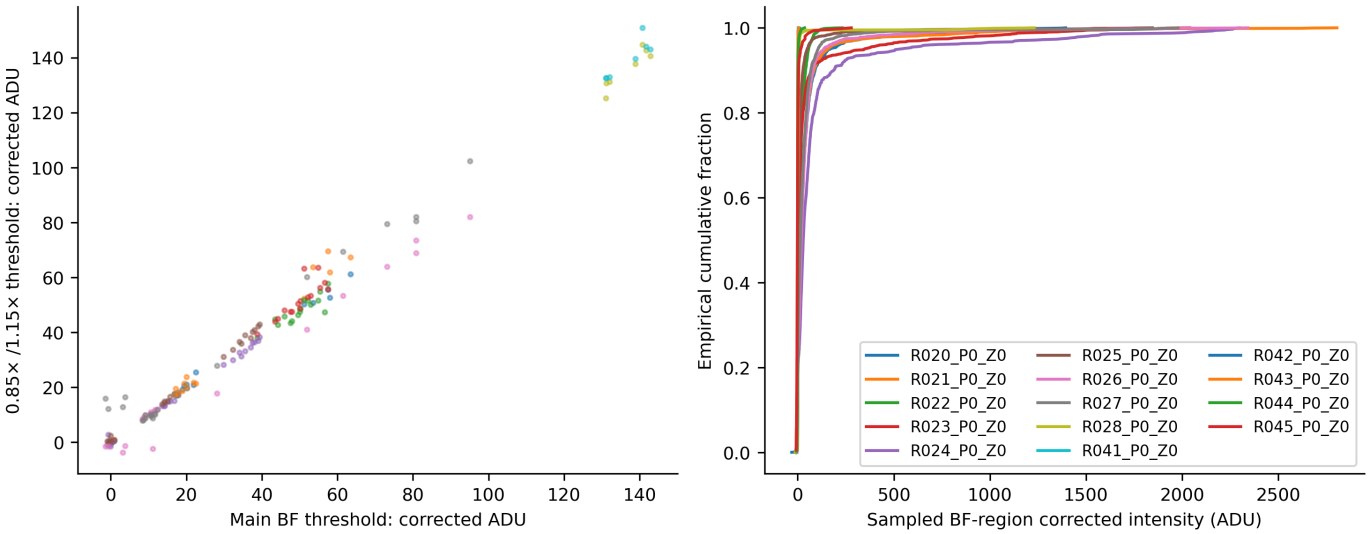
Raw static-field means are separated by batch and optical settings. All B2/B10 fluorescence Z planes use acquired ND2 event coordinates and are retained, including out-of-focus planes without a usable BF-region mask; these are optical stacks, not biological time series.

Figure 8



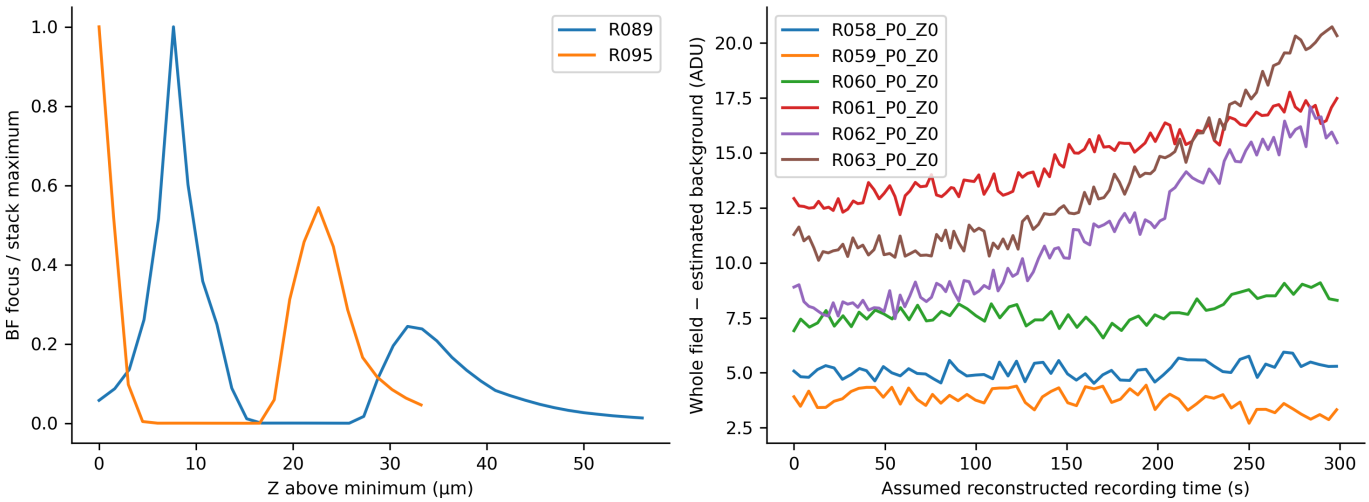
9 October S1 Cd-labelled and no-Cd exports. Clocks conflict within TIFF metadata, so the primary axis is frame index; each recording is normalized to its own initial reference.

Figure 9



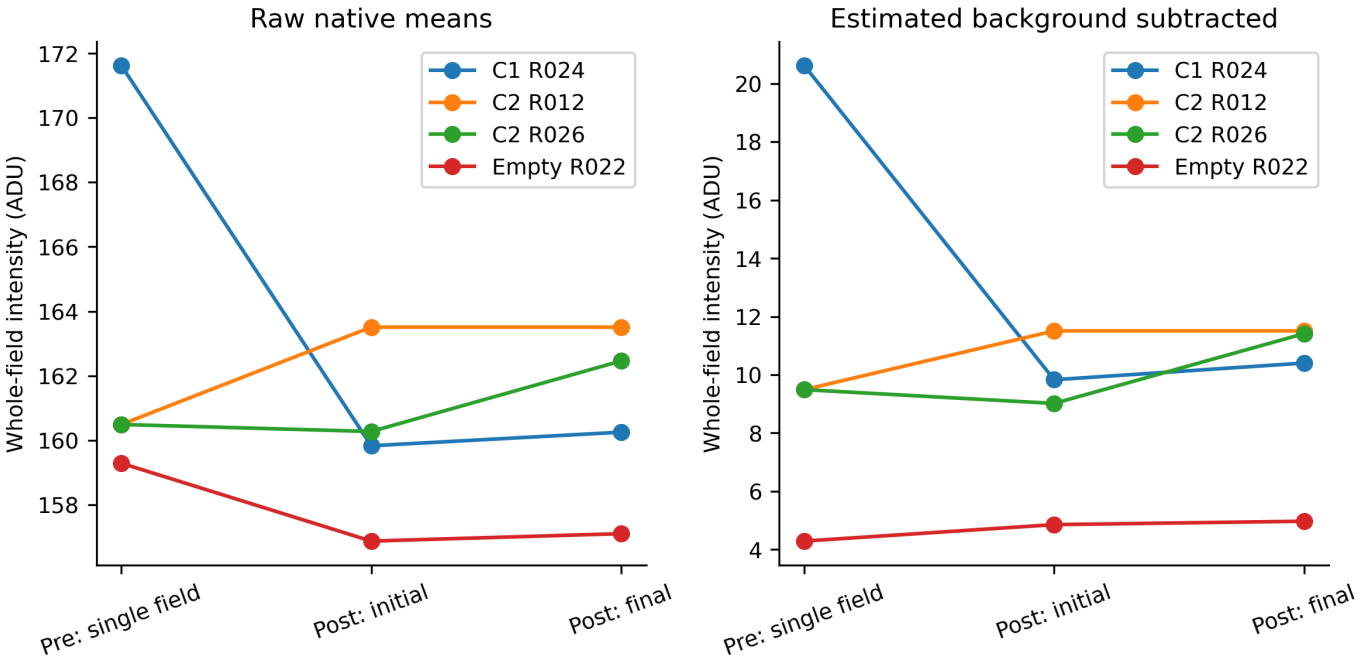
Mask-threshold sensitivity and heterogeneity of sampled BF-defined regions. Object distributions are descriptive and do not supply biological replication.

Figure 10



Z-dependent focus is an optical diagnostic. TIFF time reconstruction assumes three-position interleaving; uniform header and reconstructed clock metrics are both tabulated, while primary TIFF results remain frame-index based.

Figure 11



Candidate same-field comparisons pass stage-distance and registered BF-correlation checks. Post values use initial/final 10% medians; the C2 calcium-conflict static file has a single post frame. C1/Empty comparisons change both calcium and cadmium labels; C2 R026 keeps nominal calcium at 1 mM. Field matching does not establish same-cell identity, equal exposure or addition time.